

Lab Sheet 6 (CST-108)

Part-A (Open SSL Commands)

1. Write OpenSSL commands to do the following:
 - (a) Generate a 56-bit DES key (56-bit random number) and store it in file `des_key`
 - (b) Create a text file `Alice.txt` containing text given in Appendix-I.
 - (c) Use DES3 to encrypt `Alice.txt` and generate encrypted file `Alice.enc`.
 - (d) Now decrypt the `Alice.enc` file and store the output in `Alice.dec`
 - (e) Use Unix command “diff” to compare the `Alice.txt` file and the decrypted file `Alice.dec`. Are they both same?

[Submit the sequence of OpenSSL commands used by you and `Alice.enc` file]

2. Download a picture and stored it in .bmp format. Name it as `Sample.bmp`. Encrypt this picture file using the AES-128-ECB (Electronic Code Book) and AES-128-CBC (Cipher Block Chaining) modes. Name the encrypted files `Sample_ecb.enc` and `Sample_cbc.enc`. Now do the following so that these encrypted pictures can be treated as pictures, and can be viewed using a picture viewing software:
 - (a) For the .bmp file, the first 54 bytes contain the header information about the picture. To set `Sample_ecb.enc` and `Sample_cbc.enc` correctly, so the encrypted files can be treated as a legitimate .bmp file, replace the header of the encrypted picture with that of the original picture. You can use following unix commands to do this.

```
$ head -c 54 Sample.bmp > header
$ tail -c +55 Sample_xyz.enc > body
$ cat header body > new_xyz.bmp
```

[Replace xyz by edb and cbc]
 - (b) Display the encrypted pictures using any picture viewing software. Can you derive any useful information about the original picture from the encrypted picture? Please explain your observations.

[Submit sequence of OpenSSL commands used by you, `Sample_ecb.enc` and `Sample_cbc.enc` files and your observations about the two encrypted files.]

3. Write OpenSSL commands to do the following:
 - (a) Generate DSA private key [Use `openssl genssa`] and store it in [dsaprivatekey.pem](#)
 - (b) Now compute an SHA1 hash for the `Alice.txt` and write it in hexadecimal form to the file named `digest.txt`.
 - (c) Sign the SHA1 hash of the file using the DSA private key and write the signature out to the file `dsasign.bin`.
 - (d) Verify the signature of the file named `Alice.txt` that is contained in the file `dsasign.bin` using the SHA1 message digest algorithm and the DSA private.

[submit the OpenSSL commands to do the above and all files generated by you]

4. PEM (originally “**P**rivacy **E**nhanced **M**ail”) is the most common format for X.509 certificates. A PEM file is a text file containing one or more items in Base64 ASCII encoding, each with plain-text headers and footers. Write your own base64 encoder and decoder in C / C++.
- (a) Use your encoder to encode the file Alice.txt. Store the output in file out.pem
 - (b) Use OpenSSL command to decode the file. Write the output in file out2.txt. Are both files same?
 - (c) Use OpenSSL command to encode in.txt. Write the output in file out3.txt
 - (d) Use your decoder to decode the file. Write the output in file out4.txt. Are both files same?

Part-B

5. Execute the following programs on a machine having Intel Processor and Ubuntu OS.
[Submit output for each printf and explain them.]

```
(a) int main(){
    short port = 5120, port2;
    char *p = (char *) &port;
    char *p2 = (char *) &port2;

    for (int i = 0; i < sizeof(port); i++)
        printf("%x ", (unsigned char) *p++);
    printf("\n");

    port2 = htons(port);
    for (int i = 0; i < sizeof(port2); i++)
        printf("%x ", (unsigned char) *p2++);
}
```

```
(b) int main() {
    sigset_t set, oset, zero;
    sigemptyset(&set);
    sigemptyset(&oset);
    sigemptyset(&zero);
    sigaddset(&set, SIGCHLD);
    sigaddset(&set, SIGHUP);
    sigprocmask(SIG_BLOCK, &set, &oset);
    printf("%signal mask set\n");
    kill(getpid(), SIGCHLD);
    sigsuspend(&zero);
    sigprocmask(SIG_SETMASK, NULL, &set);
    if (sigismember(&set, SIGHUP)) printf("SIGHUP \n");
    if (sigismember(&set, SIGCHLD)) printf("SIGCHLD \n");
    if (sigismember(&set, SIGINT)) printf("SIGINT \n");
}
```

6. Modify the above program to install a signal handler for SIGCHLD. Execute the program. Write the output of the modified program along with the reason.
[Submit modified program, output and reason]

Appendix - I

Alice.txt

Down, down, down. There was nothing else to do, so Alice soon began talking again. “Dinah’ll miss me very much to-night, I should think!” (Dinah was the cat.) “I hope they’ll remember her saucer of milk at tea-time. Dinah my dear! I wish you were down here with me! There are no mice in the air, I’m afraid, but you might catch a bat, and that’s very like a mouse, you know. But do cats eat bats, I wonder?” And here Alice began to get rather sleepy, and went on saying to herself, in a dreamy sort of way, “Do cats eat bats? Do cats eat bats?” and sometimes, “Do bats eat cats?” for, you see, as she couldn’t answer either question, it didn’t much matter which way she put it. She felt that she was dozing off, and had just begun to dream that she was walking hand in hand with Dinah, and saying to her very earnestly, “Now, Dinah, tell me the truth: did you ever eat a bat?” when suddenly, thump! thump! down she came upon a heap of sticks and dry leaves, and the fall was over.

Alice was not a bit hurt, and she jumped up on to her feet in a moment: she looked up, but it was all dark overhead; before her was another long passage, and the White Rabbit was still in sight, hurrying down it. There was not a moment to be lost: away went Alice like the wind, and was just in time to hear it say, as it turned a corner, “Oh my ears and whiskers, how late it’s getting!” She was close behind it when she turned the corner, but the Rabbit was no longer to be seen: she found herself in a long, low hall, which was lit up by a row of lamps hanging from the roof.